Data Quality and Validation in ETL Assignment

Question 1: Define Data Quality in the context of ETL pipelines. Why is it more than just data cleaning?

Answer

Data quality refers intend use and high quality data is **accurate, complete, unique, clear, valid, consistent, timely** in ETL pipeline.

Bad quality data does not enter into ETL.

Cleaning remove errors (duplicates, wrong formats).

Quality ensures **validity, integrity, and usability** for business decisions.

Example: Even if nulls are removed, incorrect mappings or missing referential integrity can still harm dashboards.

Question 2: Explain why poor data quality leads to misleading dashboards and incorrect decisions.

Answer

Poor quality data leads to

Wrong total (duplicate transactions inflate revenue).

Missing value (null amounts reduce accuracy).

Incorrect joins (invalid customer IDs break reports).

Invalid product code

missing patient id

Duplicate billing record

Incorrect or inconsistent date

Question 3 : What is duplicate data? Explain three causes in ETL pipelines.

Answer

Duplicates data occurs in the multiple row the same real world entity in the dataset.

The ETL data integrate from different data sources, increase risk the repeated entries.

Example = same customer_id ,email_id

Causes in ETL pipelines

- Multiple Data Sources = Same entity exists in CRM, ERP, and marketing systems.
- Faulty Joins = Many-to-many joins multiply records unintentionally.
- Manual Data Entry Errors = Users enter the same record more than once.
- Missing or Weak Primary Keys = No unique identifier to distinguish records.
- Incremental Load Failures = Old data reloaded due to missing watermark or timestamp logic.

Question 4: Differentiate between exact, partial, and fuzzy duplicates.

Answer

- **Exact duplicate** = all column in the record completely identical.

Example: Same Name, Email, Phone, Address

- **Partial duplicate** =only key business fields match, other column different.

Example:

Same Email or Phone number

Different Name spellings or addresses

- **Fuzzy duplicate** = same entity but with slightly variation .

Examples:

"Rahul Sharma" vs "R. Sharma"
"Bengaluru" vs "Bangalore"

Question 5: Why should data validation be performed during transformation rather than after loading?

Answer

During transformation because ensure only clean validate data is loaded. during transformation prevents dirty data, efficiency.

After loading errors already polluted warehouse -costly fixes

Question 6: Explain how business rules help in validating data accuracy. Give an example.

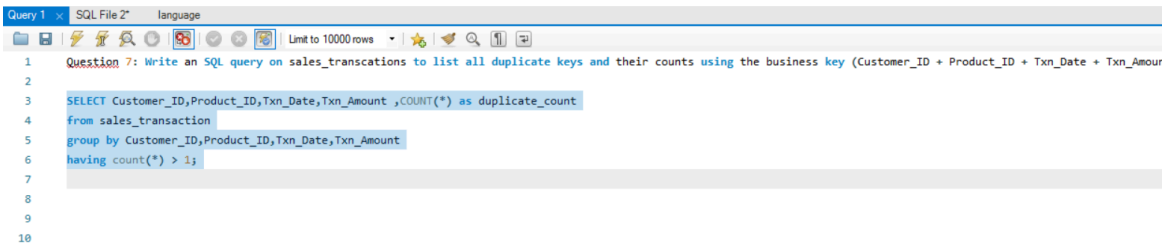
Answer

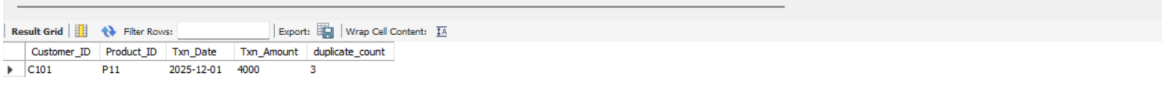
Business rules = logic defined by the organization. Example Rule: Quantity must be > 0 and Txn_Amount = Quantity × Unit Price.

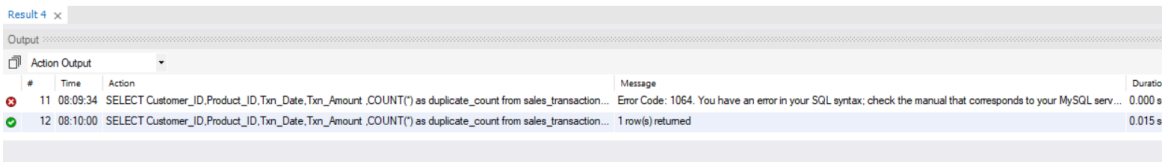
- Txn_ID 205 violates rule (Quantity = Null).
- Txn_ID 206 violates rule (Txn_Amount = Null).

Question 7: Write an SQL query on sales_transcations to list all duplicate keys and their counts using the business key (Customer_ID + Product_ID + Txn_Date + Txn_Amount).

Answer



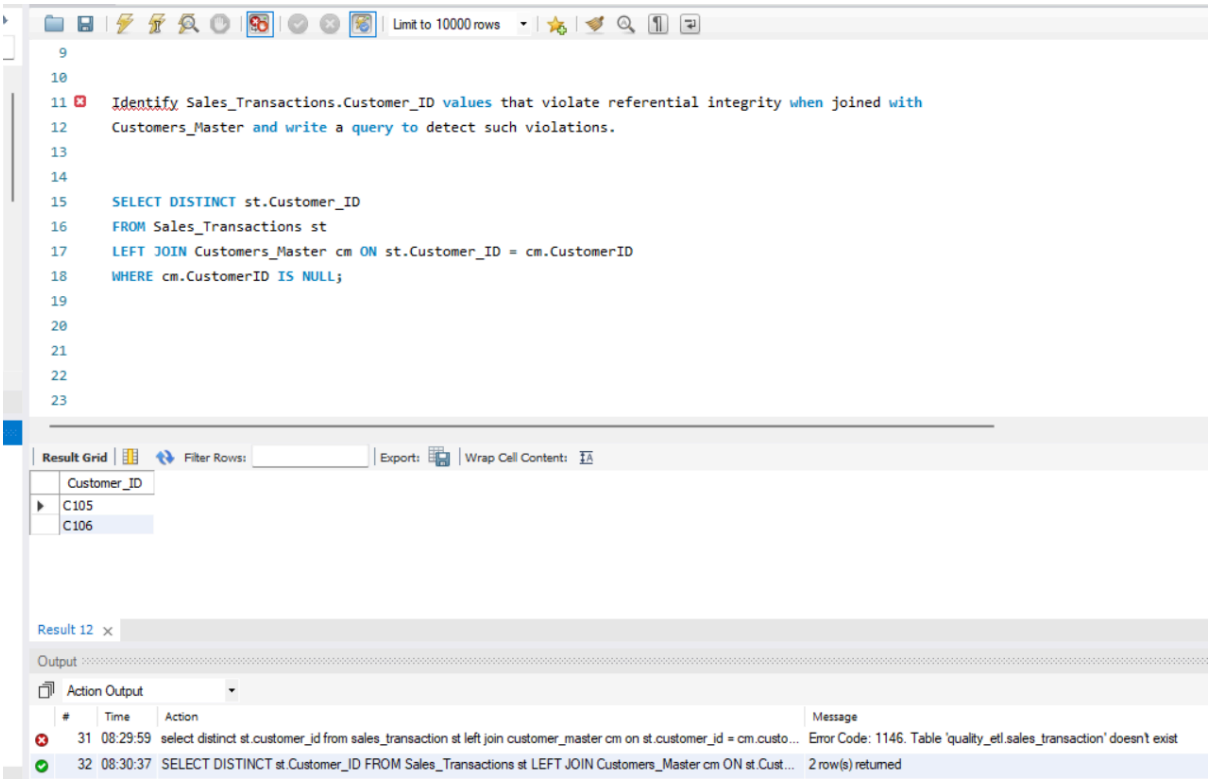




Question 8 : Enforcing Referential Integrity Assume the following customer_masters table:

Identify Sales_Transactions.Customer_ID values that violate referential integrity when joined with Customers_Master and write a query to detect such violations.4

Answer



- C105 (from Txn_ID 206)
- C106 (from Txn_ID 207)